

When Identities Collapse: A Stress-Test Benchmark for Multi-Subject Personalization

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Abstract

001 Subject-driven text-to-image diffusion models [8, 14]
002 have achieved remarkable success in preserving single
003 identities, yet their ability to compose multiple inter-
004 acting subjects remains largely unexplored and highly
005 challenging. Existing evaluation protocols typically
006 rely on global CLIP metrics [12], which are insensitive
007 to local identity collapse and fail to capture the sever-
008 ity of multi-subject entanglement. In this paper, we
009 introduce a comprehensive stress-test benchmark de-
010 signed to evaluate multi-subject personalization under
011 varying degrees of complexity, ranging from 2 to 10
012 interacting identities. To accurately measure identity
013 preservation, we shift the evaluation paradigm from
014 CLIP-based similarity to DINOv2 [10] feature match-
015 ing, and propose a novel metric, Subject Collapse Rate
016 (SCR), to explicitly quantify the proportion of iden-
017 tity loss in generated scenes. We conduct an exten-
018 sive evaluation of recent state-of-the-art models (MO-
019 SAIC [15], XVerse [4], PSR [16]) under our benchmark.
020 Our quantitative and qualitative results reveal a catas-
021 trophic failure in current models: as the number of
022 subjects increases, the SCR approaches 100%, indi-
023 cating complete identity confusion and attention leak-
024 age. Furthermore, we uncover a critical trade-off where
025 models sacrifice fine-grained identity fidelity (DINOv2)
026 to maintain macro-level text-image alignment (CLIP-
027 T). Our benchmark and metrics highlight a significant
028 gap between semantic and physical disentanglement,
029 providing a rigorous foundation for future research in
030 multi-subject generative systems.

031 1. Introduction

032 Generative AI has fundamentally transformed visual
033 content creation, with subject-driven text-to-image dif-
034 fusion models demonstrating an unprecedented ability
035 to generate customized images based on a few refer-

ence photos [6, 8, 14]. Early breakthroughs in person-
alization focused primarily on single-subject scenarios,
successfully tackling the challenge of "who appears in
the scene." Recent advancements have further extended
this capability to multi-subject composition, employing
sophisticated techniques such as layout-to-region bind-
ing [17, 19] and non-invasive text-stream modulation to
ensure that multiple referenced identities remain dis-
tinct [4, 15, 16].

Despite these rapid advancements, the field remains
constrained by a critical, yet underexplored, limitation:
how well can these models compose many identities
when they are deeply entangled through physical in-
teraction and occlusion? In real-world applications,
subjects rarely exist in isolated bounding boxes; they
hug, grapple, and occlude one another. When multiple
subjects interact, models based on architectures like
DiT [11] or FLUX [9] often suffer from severe atten-
tion leakage. Because all tokens compete in a shared
global attention space, identity cues, attributes, and
structural priors from one subject frequently bleed into
another. Consequently, what appears as a failure in
identity preservation is, at its core, a failure in physi-
cally grounded attention routing.

Compounding this issue is the inadequacy of current
evaluation metrics. The standard protocol for mea-
suring personalization quality heavily relies on CLIP-
based [12] image-to-image similarity. However, CLIP
is notoriously insensitive to fine-grained local features
and structural integrity. In a multi-subject scene, even
if a generated subject's face is completely distorted or
incorrectly swapped with another identity, CLIP may
still output an artificially high similarity score simply
because the global semantic context matches. This
metric blind spot masks the true severity of multi-
subject entanglement and collapse, creating a false
sense of progress.

To bridge this gap between semantic disentanglement and physical disentanglement, we present a systematic stress-test evaluation framework specifically

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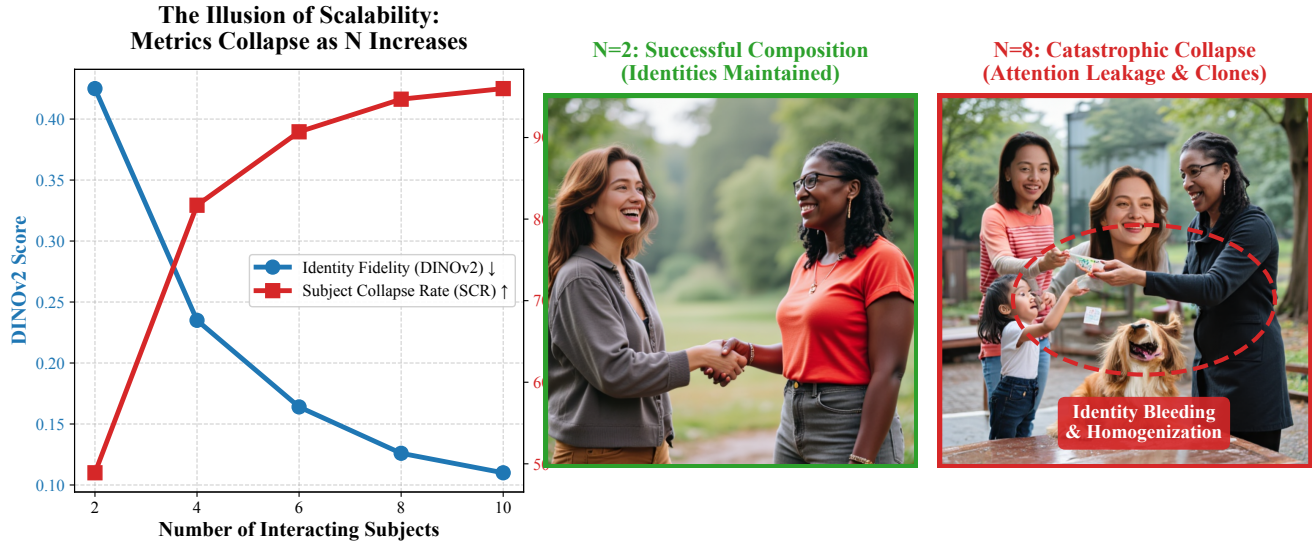


Figure 1. Catastrophic Identity Collapse in Multi-Subject Personalization. (Left) Our quantitative analysis reveals the illusion of scalability: as the number of interacting subjects (N) increases, the fine-grained identity fidelity (DINOv2) plummets, and the Subject Collapse Rate (SCR) skyrockets to $> 95\%$. (Middle) At $N = 2$, state-of-the-art models (e.g., MOSAIC [15]) successfully maintain distinct identities and structural integrity. (Right) When evaluated on our proposed stress-test benchmark at $N = 8$, models experience severe attention leakage, resulting in identity bleeding and homogenization (generating clones of a single dominant identity).

designed for extreme multi-subject composition, directly addressing the core topics of benchmark and evaluation metrics for personalization quality. Our goal is to rigorously quantify the limits of current state-of-the-art models when faced with an increasing number of interacting identities.

- In summary, our main contributions are threefold:
- **A Scalable Multi-Subject Benchmark:** We construct a comprehensive testing suite comprising prompts that scale from 2 to 10 interacting subjects, featuring various scene types including interaction, occlusion, and neutral layouts.
 - **A Rigorous Evaluation Paradigm:** We shift the identity evaluation metric from the globally-biased CLIP to the structurally-sensitive DINOv2 [10]. Furthermore, we propose the Subject Collapse Rate (SCR), a novel metric that explicitly quantifies the percentage of subjects that lose their identity in a generated scene.
 - **Insightful Failure Analysis:** We evaluate three recent state-of-the-art multi-subject models (MOSAIC [15], XVerse [4], PSR [16]). Our findings reveal a catastrophic failure mode where SCR approaches 100% as the subject count increases. We further uncover a critical trade-off: models actively sacrifice fine-grained identity fidelity (DINOv2) in favor of maintaining macro-level text alignment (CLIP-T), exposing fundamental limitations in cur-

rent attention-routing architectures.

2. Related Work

Foundations of Subject-Driven Generation. The rapid evolution of text-to-image diffusion models [13] has enabled profound new capabilities in personalized image generation. Foundational techniques like DreamBooth [14] and Textual Inversion [6] first demonstrated that a pre-trained model could learn novel subjects using only a few reference images. To improve efficiency and spatial control, subsequent works introduced parameter-efficient fine-tuning like LoRA [7], structural conditioning such as ControlNet [20], and direct image-prompt adapters like IP-Adapter [18]. While these methods achieve remarkable single-subject fidelity, they struggle with severe attribute binding and identity bleeding when tasked with generating multiple distinct subjects simultaneously.

Multi-Subject Personalization: Solving "Who is Who". Under DiT [11] and FLUX [9] backbones, early multi-subject research primarily addressed a subject-centric question: how can multiple referenced identities remain distinct under shared generation dynamics? XVerse [4] is a representative method in this direction. It converts reference images into token-specific text-stream modulation offsets, injecting subject-specific control along the text-conditioning pathway rather

130	than directly perturbing image latents. This non-	183
131	invasive design improves identity controllability while	
132	avoiding direct corruption of the shared image repre-	
133	sentation. Similarly, MOSAIC [15] tackles this prob-	
134	lem from a representation-centric perspective. By en-	
135	forcing semantic correspondence and feature disent-	
136	glement across references, it explicitly separates sub-	
137	jects in the feature space, reducing identity blending.	
138	Taken together, these methods move personalization	
139	beyond simple fidelity preservation toward explicit sub-	
140	ject separation, establishing the first layer of control-	
141	lable multi-subject generation.	
142	Layout-to-Region Binding: Solving "Where Each	
143	Subject Goes". Once subject identity is preserved,	
144	the next challenge is spatial grounding. This moti-	
145	vates a second line of work on layout-to-region bind-	
146	ing, aiming to align references, prompts, and spatial	
147	regions. ContextGen [17] introduces Contextual Lay-	
148	out Anchoring (CLA) and Identity Consistency Atten-	
149	tion (ICA) to stabilize the correspondence between in-	
150	stances and target positions. AnyMS [19] addresses the	
151	same challenge in a training-free manner by decoupling	
152	attention at global and local levels to jointly maintain	
153	prompt alignment and layout fidelity. 3DIS-FLUX [21]	
154	approaches the problem through depth-guided scene	
155	construction followed by FLUX-based rendering, uti-	
156	lizing depth as a compact structural prior. However,	
157	when the depth map underspecifies hidden geometry or	
158	front-back ordering, occlusion relations may still col-	
159	lapse. In essence, layout binding methods solve the	
160	"where" problem but only partially address the phys-	
161	ically harder question of what should remain visible	
162	under overlap.	
163	Attention Leakage and Entanglement in	
164	DiT/FLUX. To understand multi-subject failures,	
165	it is crucial to examine the mechanics of attention	
166	leakage. In DiT-style generators, global attention is	
167	both a strength and a liability. While it enables rich	
168	long-range interaction, it also allows identity cues,	
169	attributes, colors, and poses from one subject to leak	
170	into another. This issue is especially pronounced in	
171	FLUX [9], where the later single-stream blocks flatten	
172	text and image modalities into a tightly fused se-	
173	quence, potentially losing the structural inductive bias	
174	for distinct instances. From this perspective, semantic	
175	entanglement is not merely an identity-preservation	
176	failure, but an attention-routing failure. Existing	
177	methods attempt to mitigate this through various	
178	strategies—XVerse via text-stream modulation, MO-	
179	SAIC via semantic disentanglement, and AnyMS via	
180	decoupled attention flows. This progression suggests	
181	that multi-subject generation is fundamentally con-	
182	strained by how well the model can route attention	
	without cross-instance contamination.	
	Interaction and Occlusion Reasoning. The final and	
	most challenging layer is physical interaction and oc-	
	clusion reasoning. At this stage, the problem shifts	
	to who occludes whom, whether partially hidden in-	
	stances remain geometrically plausible, and whether	
	the model respects front-back ordering under over-	
	lap. SeeThrough3D [1] explicitly models which ob-	
	ject should appear in front and how partial visibility	
	should be rendered consistently, connecting naturally	
	to the notion of amodal completion. LayerBind [2]	
	leverages the coarse-to-fine denoising dynamics of DiTs,	
	establishing layout and visible order at early steps	
	through layered initialization. Nevertheless, deeply en-	
	tangled interactions—such as hugging, grappling, or in-	
	terleaved articulated bodies—remain exceptionally dif-	
	ficult. The challenge lies not only in visibility order-	
	ing but also in fine-grained contact geometry, mutual	
	deformation, and physically consistent completion un-	
	der occlusion. Our work explicitly targets this gap,	
	providing a rigorous benchmark and novel evaluation	
	metrics to measure where current models fail in physi-	
	cally grounded interaction reasoning, pushing forward	
	the frontier of multi-subject personalization.	
	3. Multi-Subject Benchmark	
	To systematically evaluate the limits of current subject-	
	driven diffusion models in handling complex multi-	
	entity compositions, we construct a rigorous, scalable	
	stress-test benchmark. Unlike existing datasets that	
	primarily focus on single or dual subjects in isolated	
	settings, our benchmark is specifically designed to as-	
	sess identity preservation and physical interaction rea-	
	soning as the number of subjects scales from 2 to 10.	
	This directly addresses the critical need for dataset cu-	
	ration for benchmarking personalized generative mod-	
	els.	
	3.1. Subject Pool Construction	
	We establish a diverse and challenging subject pool	
	by sourcing reference images from two well-established	
	personalization datasets: the XVerse dataset [4] and	
	the COSMISC dataset [15]. These datasets provide a	
	wide array of human and animal identities with vary-	
	ing attributes, poses, and clothing. We curate a uni-	
	fied subject pool by extracting high-quality reference	
	images and assigning a unique identifier (e.g., S001,	
	S002) to each distinct identity. This unified pool en-	
	sures that the models are tested on a consistent set of	
	challenging visual features.	

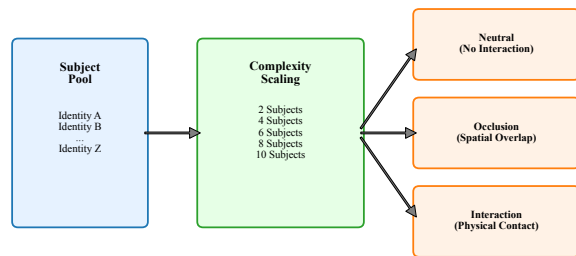


Figure 2. Multi-Subject Benchmark Construction. Our pipeline samples identities from a unified subject pool (left) to populate prompts of increasing complexity (2 to 10 subjects). Prompts are systematically categorized into Neutral, Occlusion, and Interaction scenarios (right) to isolate and test specific failure modes in attention routing and geometric reasoning.

3.2. Benchmark Design and Scalability

The core of our benchmark is a set of carefully crafted evaluation prompts designed to test both scalability and spatial reasoning. We define five distinct difficulty levels based on the subject count: 2, 4, 6, 8, and 10 subjects, as illustrated in Figure 2.

For each subject level, we design 15 unique prompts, resulting in a total of 75 base prompts. To thoroughly investigate the impact of spatial entanglement on attention leakage, we strictly control the prompt IDs (1-75) and categorize them into three distinct scene types, ensuring an even distribution of 5 prompts per type within each subject level:

- Neutral (No Interaction): Subjects are placed in the same scene without physical overlap (e.g., "Subject A and Subject B standing next to each other"). These prompts test the model's baseline ability to prevent identity bleeding across spatially separated regions.
- Occlusion: One subject partially blocks the view of another (e.g., "Subject A standing behind Subject B"). These prompts test the model's amodal completion and depth ordering capabilities.
- Interaction: Subjects are physically engaged with one another (e.g., "Subject A hugging Subject B" or "Subject A hugging Subject B while Subject C stands behind Subject D"). These are the most challenging prompts, testing the model's ability to maintain fine-grained contact geometry and identity under severe attention entanglement.

3.3. Evaluation Protocol

We evaluate three recent state-of-the-art multi-subject personalization models: XVerse [4], MOSAIC [15], and

PSR [16]. These models were specifically selected because they represent the frontier of non-invasive, attention-modulation techniques designed to overcome the limitations of naive region-masking, making them the strongest candidates for complex physical interactions. For each of the 75 prompts, the models take the text prompt and the corresponding subject reference images as input. To account for the stochastic nature of diffusion models, we generate images using 3 random seeds per prompt. This yields a total of 225 generated images per model (675 images across all three models), providing a statistically robust foundation for our quantitative analysis. All generation metadata, including prompt IDs, subject assignments, and output paths, are strictly tracked to ensure reproducible evaluation.

4. Evaluation Metrics

A central contribution of our benchmark is identifying the limitations of standard personalization metrics when applied to multi-subject composition. Existing protocols heavily rely on global CLIP embeddings [12]. However, global embeddings are highly tolerant of local structural distortions and identity bleeding, rendering them inadequate for measuring severe identity entanglement. To provide a rigorous and comprehensive assessment that aligns with the need for advanced evaluation metrics for personalization quality, we introduce a multi-tiered evaluation framework.

4.1. Text-Image Alignment (CLIP-T)

To measure how well the generated image aligns with the semantic intent of the text prompt, we compute the cosine similarity between the CLIP [12] text embedding of the prompt and the CLIP image embedding of the generated output.

While CLIP-T is a standard metric for semantic fidelity, we observe a critical caveat in multi-subject scenarios: models may achieve high CLIP-T scores by generating a generic group of people that matches the macro-semantics of the prompt (e.g., "a group of people"), while completely failing to preserve the specific identities requested. Therefore, CLIP-T must be analyzed in conjunction with fine-grained identity metrics.

4.2. Identity Preservation: From CLIP to DINOv2

Traditionally, identity preservation is measured by computing the cosine similarity between the CLIP image embedding of the generated subject and the reference image (denoted as CLIP-I). However, our experiments reveal that CLIP-I scores remain artificially high even when subjects undergo severe identity bleeding or facial distortion.

To capture these local structural failures, we shift our primary identity evaluation from CLIP to DINOv2 [10]. Unlike CLIP, which is trained primarily via contrastive language-image pre-training and tends to prioritize global semantic layout (e.g., "a person with glasses"), DINOv2 is a self-supervised vision transformer trained via image-level and patch-level objectives. This training paradigm grants DINOv2 an exceptional sensitivity to fine-grained local features, part-level correspondence, and structural geometry, making it significantly more robust for evaluating identity preservation under complex physical interactions where semantic features might otherwise bleed. While earlier versions like DINOv1 [?] also capture structural priors, DINOv2 leverages a significantly larger and curated pre-training dataset along with an improved objective function, resulting in more discriminative and stable feature embeddings for complex multi-entity scenes.

While a rigorous subject-level evaluation would ideally require instance segmentation masks to isolate each generated subject, such masks are notoriously difficult to obtain accurately in highly entangled multi-subject scenes (e.g., severe occlusion or physical interaction). Therefore, as an established proxy for overall identity fidelity, we compute the DINOv2 image embedding of the entire generated image I_{gen} and calculate its average cosine similarity against the embeddings of all N individual reference images $\{I_{ref}^{(1)}, I_{ref}^{(2)}, \dots, I_{ref}^{(N)}\}$:

$$\text{DINOv2 Score} = \frac{1}{N} \sum_{i=1}^N \cos(\text{DINOv2}(I_{gen}), \text{DINOv2}(I_{ref}^{(i)})) \quad (1)$$

Although comparing a multi-subject scene embedding against single-subject reference embeddings introduces scene complexity into the score, this formulation serves as a highly effective penalty mechanism. It strictly demands that the structural identity of every requested subject be strongly represented in the global feature space.

4.3. Subject Collapse Rate (SCR)

Average similarity scores can obscure catastrophic failures of individual subjects within a complex scene. For instance, in an 8-subject image, if 7 subjects are perfectly generated but 1 subject is completely missing or morphed into another identity, the mean DINOv2 score might still appear acceptable.

To explicitly quantify these localized failures, we propose the Subject Collapse Rate (SCR), conceptually illustrated in Figure 3. While this metric still uti-

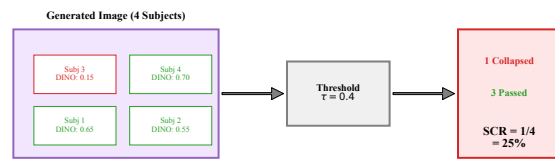


Figure 3. Subject Collapse Rate (SCR). Unlike average similarity scores which mask individual failures, SCR explicitly counts the proportion of subjects whose DINOv2 identity similarity falls below a strict threshold τ . This provides a more realistic measure of multi-subject entanglement.

lizes the scene-level generated embedding, it shifts the evaluation from a continuous average to a strict discrete thresholding. We define a subject as "collapsed" if its DINOv2 cosine similarity with the reference image falls below a predefined threshold τ . The SCR for a given generated image is defined as the ratio of collapsed subjects to the total number of subjects:

$$\text{SCR}_{@ \tau} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\cos(\text{DINOv2}(I_{gen}), \text{DINOv2}(I_{ref}^{(i)})) < \tau] \quad (2)$$

where $\mathbb{1}[\cdot]$ is the indicator function. Because DINOv2 similarities typically occupy a lower and more discriminative numerical range than CLIP, we employ strict thresholds $\tau \in \{0.4, 0.5, 0.6\}$, which empirical visual inspection confirms align with severe human-perceivable identity loss. Importantly, this metric naturally penalizes the "Homogenization" failure mode: if a model generates multiple clones of a single dominant subject, only that subject's reference will yield a high similarity, while the remaining $N - 1$ subjects will fall below the threshold, correctly driving the SCR towards 1.0. A lower SCR indicates better multi-subject preservation, while an SCR approaching 1.0 signifies a complete collapse of personalization.

5. Experiments

In this section, we present the quantitative and qualitative results of our evaluation. We benchmark three state-of-the-art multi-subject personalization models: MOSAIC [15], XVerse [4], and PSR [16], across five difficulty levels ranging from 2 to 10 interacting subjects.

5.1. Quantitative Results: The Limits of Personalization

To rigorously assess model performance, we first examine the overall trend of identity preservation as the subject count increases. The quantitative results, averaged across all scene types, are summarized in Table 1 and visualized in Figure 4.

Catastrophic Identity Forgetting The most striking finding from our benchmark is the catastrophic collapse of identity fidelity when scaling beyond 4 subjects. As shown in the DINOv2 scores, all models experience a precipitous drop. For instance, MOSAIC [15], which performs best at the 2-subject level with a DINOv2 score of 0.425, plummets to 0.110 at 10 subjects. XVerse [4] and PSR [16] follow a nearly identical trajectory, dropping from 0.355 and 0.325 to 0.104 and 0.101, respectively.

This failure is further magnified when examining the Subject Collapse Rate (SCR). At the most lenient threshold ($\tau = 0.4$), MOSAIC exhibits a 48.9% collapse rate even with just 2 subjects. When pushed to 8 subjects, the SCR skyrockets to 94.7% for MOSAIC, 94.4% for XVerse, and 95.0% for PSR. At 10 subjects, the SCR for all models exceeds 96%, indicating that nearly every generated subject has lost its intended identity. This quantitative evidence strongly supports our hypothesis: current attention-routing mechanisms are fundamentally inadequate for resolving dense physical entanglement.

Model Comparison While all models fail at extreme subject counts, MOSAIC demonstrates a noticeably stronger baseline in low-complexity scenarios. At 2 subjects, MOSAIC achieves the highest DINOv2 score (0.425) and the lowest SCR (48.9%), outperforming XVerse (0.355 / 58.9%) and PSR (0.325 / 63.3%). This multifaceted performance profile is further detailed in Figure 5, which compares the semantic, style, structure, and identity metrics simultaneously. This suggests that MOSAIC’s representation-centric disentanglement provides a more robust defense against attention leakage when the scene is relatively sparse. However, this advantage quickly dissipates as the subject count reaches 6 and beyond, underscoring the universal scalability bottleneck in current DiT/FLUX architectures [9, 11].

5.2. The Trade-off: Semantic vs. Physical Disentanglement

A counter-intuitive phenomenon emerges when we analyze the Text-Image Alignment (CLIP-T) [12] scores

alongside the identity metrics. While DINOv2 [10] scores plummet as the subject count increases, the CLIP-T scores for all models actually exhibit a slight upward trend. For example, PSR’s CLIP-T score rises from 0.274 (2 subjects) to 0.309 (10 subjects), and MOSAIC rises from 0.261 to 0.300.

This exposes a critical trade-off mechanism inherent in current diffusion models. When faced with the overwhelming complexity of generating 8 or 10 distinct, interacting individuals, the models actively “take a shortcut.” Instead of attempting to faithfully reconstruct each specific identity—which would require precise, uncorrupted attention routing—the models revert to generating a generic “group of people” that satisfies the macro-level semantic constraints of the prompt. Consequently, the global CLIP-T score improves, but at the total expense of personalized identity.

5.3. Qualitative Failure Analysis

To better understand how these models fail, we perform a qualitative visual analysis of the generated scenes.

Visual inspection corroborates our quantitative findings and reveals three primary failure modes during multi-subject interaction:

1. **Identity Bleeding:** Features from one subject (e.g., clothing color, facial structure, glasses) seamlessly bleed into an adjacent subject, especially during physical contact (e.g., hugging).
2. **Homogenization:** In dense scenes (8-10 subjects), the model often generates multiple copies of the same dominant reference identity, completely ignoring the other requested subjects.
3. **Amodal Collapse:** When subjects are occluded, the model frequently fails to render the hidden geometry correctly, resulting in fused limbs or disembodied floating heads.

6. Discussion

Our comprehensive evaluation uncovers several fundamental bottlenecks in current multi-subject personalization models, pointing to critical areas for future research within the generative AI community.

6.1. The "Semantic Shortcut" Hypothesis

The divergent trends between CLIP-T [12] (which slightly improves) and DINOv2 [10] (which drastically drops) as subject count increases suggest that models are not merely “failing” to generate images, but are actively optimizing for the wrong objective. In DiT [11] and FLUX [9] architectures, when the global attention mechanism is overwhelmed by multiple, competing identity embeddings, the model defaults to the

Table 1. Quantitative Evaluation on Multi-Subject Personalization Benchmark. We report CLIP-T, CLIP-I, DINOv2, and Subject Collapse Rate (SCR@0.4) across varying subject counts (2 to 10). As subject count increases, DINOv2 sharply declines and SCR approaches 100%, indicating catastrophic identity failure.

Subjects	MOSAIC [15]				XVerse [4]				PSR [16]			
	CLIP-T $\uparrow$	CLIP-I $\uparrow$	DINOv2 $\uparrow$	SCR $\downarrow$	CLIP-T $\uparrow$	CLIP-I $\uparrow$	DINOv2 $\uparrow$	SCR $\downarrow$	CLIP-T $\uparrow$	CLIP-I $\uparrow$	DINOv2 $\uparrow$	SCR $\downarrow$
2	0.261	0.695	0.425	48.9%	0.268	0.646	0.355	58.9%	0.274	0.647	0.325	63.3%
4	0.289	0.586	0.235	81.7%	0.279	0.593	0.211	80.0%	0.292	0.568	0.189	85.6%
6	0.297	0.535	0.164	90.7%	0.275	0.550	0.142	93.0%	0.302	0.537	0.136	94.1%
8	0.302	0.517	0.126	94.7%	0.276	0.532	0.123	94.4%	0.304	0.517	0.117	95.0%
10	0.300	0.504	0.110	96.0%	0.283	0.524	0.104	96.4%	0.309	0.500	0.101	97.8%

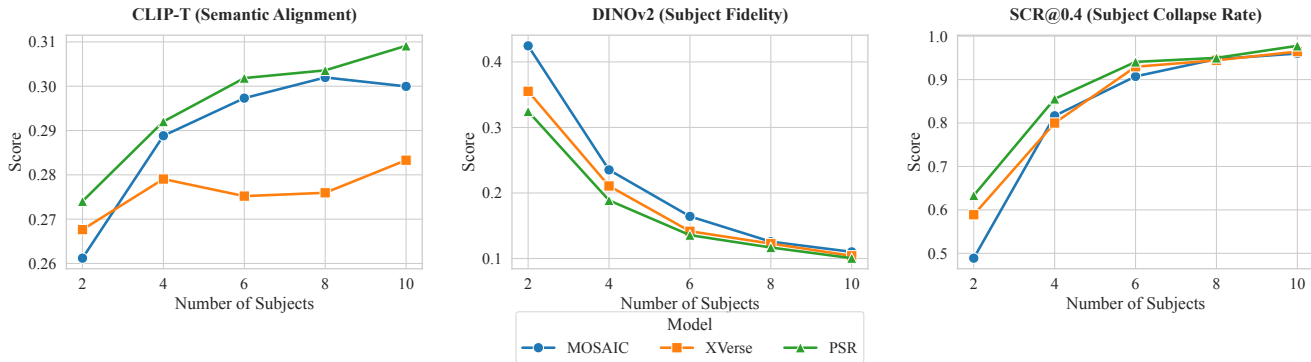


Figure 4. Performance Trends across Subject Counts. (Left) DINOv2 identity similarity exhibits a precipitous drop for all models as scenes become denser. (Right) Subject Collapse Rate (SCR@0.4) skyrockets from ~50% at 2 subjects to nearly 100% at 8-10 subjects, highlighting a fundamental scalability bottleneck.

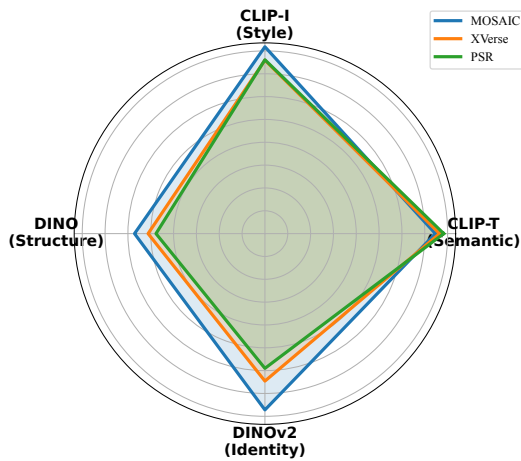


Figure 5. Comprehensive Metric Radar (2 Subjects). A multi-dimensional comparison of MOSAIC, XVerse, and PSR. While all models maintain high semantic alignment (CLIP-T), MOSAIC shows a distinct advantage in structural (DINO) and fine-grained identity (DINOv2) preservation.

to satisfy the text encoder, completely washing out the localized, high-frequency details required for identity preservation. This highlights a dangerous flaw in relying solely on CLIP for generative evaluation.

6.2. Physical Disentanglement vs. Semantic Disentanglement

While recent methods like MOSAIC [15] and XVerse [4] have made significant strides in semantic disentanglement (e.g., ensuring "dog" features don't mix with "cat" features in latent space), our benchmark reveals that they still fail at physical disentanglement. When subjects interact (e.g., hugging) or occlude one another, the boundary between their spatial regions becomes blurred. Because current models lack explicit 3D reasoning or amodal completion mechanisms [1], the physical overlap directly translates into attention leakage in the 2D feature maps. Solving "who is who" is insufficient if the model cannot resolve "who is in front of whom." Future directions could involve incorporating direct geometric priors into transformer backbones, as demonstrated by recent breakthroughs in feed-forward 3D networks like VGGT [5], temporal-spatial consistency models like NWM [3], and highly expressive rendering primitives such as Student Splatting and Scooping [22], to enforce physical boundaries and temporal-

strongest available prior: the global text prompt. We term this the "semantic shortcut." The model synthesizes a structurally plausible scene of "many people"

spatial consistency during the denoising process.

6.3. Limitations

We acknowledge certain limitations in our current benchmark. First, our evaluation is primarily focused on human and animal subjects; evaluating multi-object compositions involving inanimate objects with rigid geometries may yield different failure modes. Second, our scene types (neutral, occlusion, interaction) are defined via prompt engineering rather than explicit 3D layout controls [17, 19], meaning the severity of occlusion is left to the model’s interpretation.

7. Conclusion

In this paper, we presented a rigorous stress-test benchmark to evaluate the limits of subject-driven diffusion models in multi-entity compositions, directly addressing the critical need for robust evaluation metrics in the personalization domain. We demonstrated that standard CLIP-based metrics are dangerously blind to local identity collapse, and we proposed the Subject Collapse Rate (SCR) based on DINOv2 to quantify this failure. Our extensive evaluation of SOTA models (MOSAIC [15], XVerse [4], PSR [16]) yielded a sobering conclusion: while models can handle 2-4 isolated subjects, they suffer from catastrophic identity forgetting—with SCR approaching 100%—when forced to compose 6 to 10 interacting identities. This failure exposes a fundamental flaw in current global attention routing mechanisms, which actively sacrifice localized physical fidelity to satisfy macro-level semantic alignment. To overcome this bottleneck, future architectures must move beyond global token modulation and explore explicit representation disentanglement or part-level correspondence mechanisms. We hope our benchmark and the proposed SCR metric will serve as a clear, demanding target for the next generation of physically-grounded personalization models.

References

- [1] Vaibhav Agrawal, Rishubh Parihar, Pradhaan S Bhat, Ravi Kiran Sarvadevabhatla, and R Venkatesh Babu. Seethrough3d: Occlusion aware 3d control in text-to-image generation. arXiv preprint arXiv:2602.23359, 2026. 3, 7
- [2] Anonymous. Layerbind: Coarse-to-fine layered initialization for multi-subject generation. Under Review, 2025. 3
- [3] Amir Bar, Gaoyue Zhou, Danny Tran, Trevor Darrell, and Yann LeCun. Navigation world models. arXiv preprint arXiv:2412.03572, 2024. 7
- [4] Bowen Chen, Mengyi Zhao, Haomiao Sun, Li Chen, Xu Wang, Kang Du, and Xinglong Wu. Xverse: Consistent multi-subject control of identity and semantic attributes via dit modulation. arXiv preprint arXiv:2506.21416, 2025. 1, 2, 3, 4, 5, 6, 7, 8
- [5] Minghao Chen, Nikita Karaev, Andrea Vedaldi, Christian Rupprecht, and David Novotny. Vggt: Visual geometry grounded transformer. arXiv preprint arXiv:2503.11651, 2025. 7
- [6] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H Bermano, Gal Chechik, and Daniel Cohen-Or. An image is worth one word: Personalizing text-to-image generation using textual inversion. In International Conference on Learning Representations, 2023. 1, 2
- [7] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. In International Conference on Learning Representations, 2022. 2
- [8] Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. Multi-concept customization of text-to-image diffusion. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 1931–1941, 2023. 1
- [9] Black Forest Labs. Flux.1. <https://blackforestlabs.ai/>, 2024. 1, 2, 3, 6
- [10] Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, et al. Dinov2: Learning robust visual features without supervision. Transactions on Machine Learning Research, 2024. 1, 2, 5, 6
- [11] William Peebles and Saining Xie. Scalable diffusion models with transformers. In Proceedings of the IEEE/CVF International Conference on Computer Vision, pages 4199–4209, 2023. 1, 2, 6
- [12] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In International conference on machine learning, pages 8748–8763. PMLR, 2021. 1, 4, 6
- [13] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pages 10684–10695, 2022. 2
- [14] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 22500–22510, 2023. 1, 2
- [15] Dong She, Siming Fu, Mushui Liu, Qiaoqiao Jin, Hualiang Wang, Mu Liu, and Jidong Jiang. Mosaic: Multi-subject personalized generation via correspondence-aware alignment and disentanglement.

- 627 arXiv preprint arXiv:2509.01977, 2025. [1](#), [2](#), [3](#), [4](#), [5](#), [6](#),
628 [7](#), [8](#)
- 629 [16] Shulei Wang, Longhui Wei, Xin He, Jianbo Ouyang,
630 Hui Lu, Zhou Zhao, and Qi Tian. Psr: Scal-
631 ing multi-subject personalized image generation with
632 pairwise subject-consistency rewards. arXiv preprint
633 arXiv:2512.01236, 2025. [1](#), [2](#), [4](#), [5](#), [6](#), [7](#), [8](#)
- 634 [17] Ruihang Xu, Dewei Zhou, Fan Ma, and Yi Yang.
635 Contextgen: Contextual layout anchoring for identity-
636 consistent multi-instance generation. arXiv preprint
637 arXiv:2510.11000, 2025. [1](#), [3](#), [8](#)
- 638 [18] Hu Ye, Jun Zhang, Sibor provisions Liu, Xiao Han,
639 and Wei Yang. Ip-adapter: Text compatible image
640 prompt adapter for text-to-image diffusion models.
641 arXiv preprint arXiv:2308.06721, 2023. [2](#)
- 642 [19] Binhe Yu, Zhen Wang, Kexin Li, Yuqian Yuan, Wen-
643 Qiao Zhang, Long Chen, Juncheng Li, Xiao Jun, and
644 Yueting Zhuang. Anyms: Bottom-up attention decou-
645 pling for layout-guided and training-free multi-subject
646 customization. arXiv preprint arXiv:2512.23537, 2025.
647 [1](#), [3](#), [8](#)
- 648 [20] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala.
649 Adding conditional control to text-to-image diffusion
650 models. In Proceedings of the IEEE/CVF Interna-
651 tional Conference on Computer Vision, pages 3836–
652 3847, 2023. [2](#)
- 653 [21] Dewei Zhou, Ji Xie, Zongxin Yang, and Yi Yang. 3dis-
654 flux: simple and efficient multi-instance generation
655 with dit rendering. arXiv preprint arXiv:2501.05131,
656 2025. [3](#)
- 657 [22] Jialin Zhu, Jiangbei Yue, Feixiang He, and He Wang.
658 3d student splatting and scooping. arXiv preprint
659 arXiv:2503.10148, 2025. [7](#)

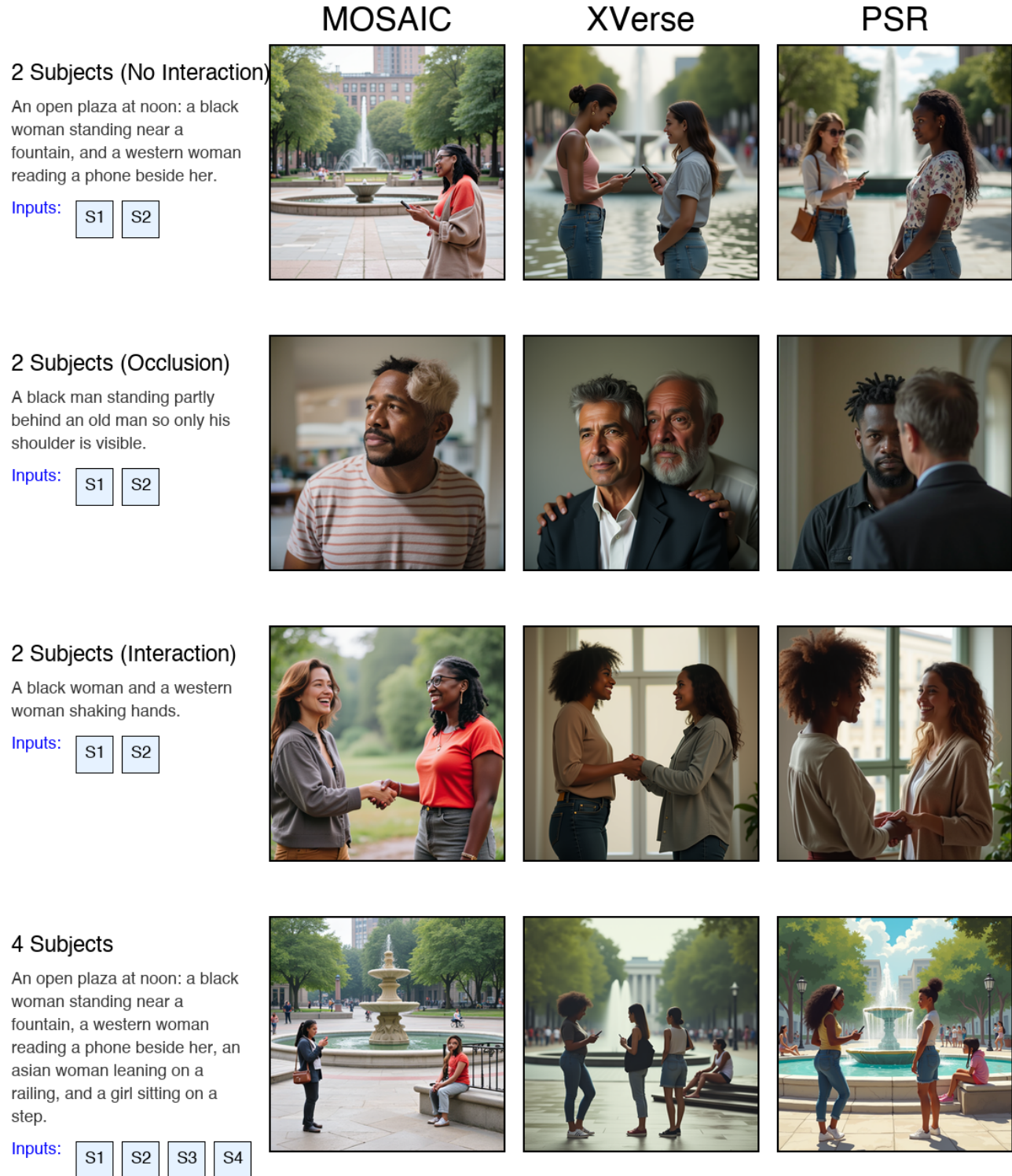


Figure 6. Qualitative Failure Analysis. Comparison between different models across various subject counts and scene types. In denser scenes (4+ subjects) or complex interactions, models exhibit severe (1) Identity Bleeding, where features merge; (2) Homogenization, generating clones of a single dominant identity; and (3) Amodal Collapse, resulting in malformed geometry under occlusion.



Prompt: A black woman and a western woman shaking hands.

Figure 7. Detailed Case Analysis of Identity Bleeding. (Left) Reference images of two distinct subjects. (Right) Generation results for a complex interaction prompt ("A black woman and a western woman shaking hands"). While the MOSAIC baseline successfully maintains identity boundaries (marked with green check), XVerse and PSR suffer from severe identity bleeding, where the features of Subject A contaminate the spatial region of Subject B (marked with red cross).